

Zhang Yifan (Jenny)

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Education

Ph.D. in Psychometrics

2022 - 2026 (*expected*)

Faculty of Education, The University of Hong Kong

Topic: Latent Variable Modeling with Bayesian Learning

M.Sc. in Data Science

2020 - 2022

Department of Statistics & Department of Computer Science, The University of Hong Kong

Capstone project: Statistical Inference for Tensor Data

B.Sc. in Information and Computation Science

2016 - 2020

Department of Mathematics, Northeastern University, China

Thesis: Analysis of Breast Cancer Prediction Based on Neural Network

Research Interests

Psychometrics; Latent Variable Modeling; Bayesian Methods; Educational Measurement; Structural Equation Modeling; Interpretable Machine Learning; AI for Measurement; Statistical Learning

Publications

Peer-Reviewed Journal Articles

Zhang, Y., & Chen, J. (2026). Alternative Bayesian Regularizations for Factor Correlations and Loadings: Lasso, Spike-and-slab, and Horseshoe *Under Review*

Chen, J., & **Zhang, Y.** (2025). Research Design and Model Estimation Under the Partially Confirmatory Latent Variable Modeling Framework with Multi-Univariate Bayesian Lassos. *Structural Equation Modeling: A Multidisciplinary Journal*, 32(2), 200–214. [doi]

Zhang, Y., & Chen, J. (2024). Accommodating and Extending Various Models for Special Effects Within the Generalized Partially Confirmatory Factor Analysis Framework. *Applied Psychological Measurement*, 48(4-5), 208-229. [doi]

Conference Presentations

Zhang, Y., & Chen, J. (2026, July). How should we model factor structures with partial knowledge? To be Presented at the *International Meeting of the Psychometric Society (IMPS)*, Seoul, Korea.

Zhang, Y., & Chen, J. (2026, April). Bayesian Graphical Models in Factor Analysis: A Prior Comparison. To be Presented at the *National Council on Measurement in Education (NCME)*, Los Angeles, California.

Zhang, Y., & Chen, J. (2025, July). Bayesian Graphical Models for Factorial Correlation Estimation. Oral presentation at the *International Meeting of the Psychometric Society (IMPS)*, Minneapolis, MN.

Zhang, Y., & Chen, J. (2025, April). A Comprehensive Comparison of Partially Confirmatory Methods in SEM. Poster presentation at the *National Council on Measurement in Education (NCME)*, Denver, CO.

Zhang, Y., & Chen, J. (2023, March). Accommodating and Extending Various Models for Special Effects within the Generalized Partially Confirmatory Factor Analysis Framework. Oral presentation at the *National Council on Measurement in Education (NCME)*, Chicago, IL. (Virtual)

Zhang, Y., & Chen, J. (2023, March). Improving Recommender System with the Partially Confirmatory Approach and Psychological Factors. Oral presentation at the *National Council on Measurement in Education (NCME)*, Chicago, IL. (Virtual)

Research Project

Statistical Inference for Tensor Data

2021

Master Capstone Project

Hong Kong, China

- Developed deep learning models for spatiotemporal sequence prediction, implementing architectures such as SRVP, ConvLSTM, Conv-TTLSTM, Deep RNN, and PredRNN.
- Conducted comparative experiments on the KTH human action dataset, evaluating model performance using PSNR, SSIM, and LPIPS metrics.
- Identified model trade-offs between prediction accuracy and computational efficiency, providing insights for selecting architectures in video prediction tasks.

Breast Cancer Prediction Based on Neural Network

2020

Undergraduate Thesis

Shenyang, China

- Analyzed over 27,000 genetic features from breast cancer datasets and performed dimensionality reduction using signal-to-noise ratio techniques.
- Developed a neural network classification framework (SDAE-LM-LVQ) to improve cancer prediction accuracy.
- Achieved 75% classification accuracy, outperforming baseline machine learning models.

Artificial intelligence training camp - machine learning

2019

Summer Camp

Shenyang, China

- Built convolutional neural network models for image classification tasks using Python and deep learning frameworks.
- Implemented end-to-end pipelines including data preprocessing, model training, and performance evaluation.

Athletic Games Information Administration System

2018

Summer Camp

Shenyang, China

- Designed front-end interfaces for a web-based athletic event management system to support event information display and user interaction.
- Collaborated with backend and database developers to integrate front-end functionality with system data and application logic.

Working Experience

The University of Hong Kong

January - June, 2024

RPG Student Coordinator

Hong Kong, China

- Represented postgraduate students in organizing the Research Postgraduate Conference, coordinating academic and social events for MPhil, PhD, and EdD students.

The University of Hong Kong

June - August, 2022

Research Assistant I

Hong Kong, China

- Applied latent variable modeling (PCFA) to machine learning contexts using large-scale recommendation data, improving model interpretability while maintaining predictive performance.
- Demonstrated how psychometric modeling enhances transparency compared with black-box recommender models; results presented at NCME 2023.

The University of Hong Kong
Research Assistant II

June - August, 2021
Hong Kong, China

- Prepared teaching materials, assignments, and assessment content for statistics and psychometrics courses.
- Used R and RMarkdown to generate reproducible course documents and visualize structural equation modeling (SEM) concepts.

Biden Consulting Company
Campus Representative

March - November, 2019
Shenyang, China

- Led campus outreach and marketing initiatives to promote company products and services.
- Coordinated alumni association operations and organized campus engagement events.
- Recognized as Outstanding Professional Manager, ranking 4th out of 100 nationwide.

Manulife Financial Centre
Asset Management Intern

February, 2019
Hong Kong, China

- Analyzed Asian market trends for asset planning, tracked financial service industry dynamics, collected data, and wrote business proposals for decision-making.
- Conducted portfolio analysis through real-time simulation for corporate asset management.
- Received Best Team Award of The 2019 Manulife GP programme.

Teaching Experience

Teaching Assistant

The University of Hong Kong

EDUR6021 , Quantitative Research Methods II	<i>Spring 2024, Spring 2025</i>
EDUR7103 , Structural Equation Modeling I	<i>Spring 2024</i>
EDUR7102 , Statistical and Psychometric Analysis with R	<i>Fall 2023</i>
EDUR7109 , Factor Analysis	<i>Fall 2023</i>
MEDD8904 , Introduction to Factor Analysis and Structural Equation Modeling	<i>Spring 2023</i>
MEDD8815 , Introduction to Statistical Methods	<i>Fall 2022</i>
MLIM6025 , Methods of Research and Enquiry	<i>Spring 2022</i>

Selected Awards & Honors

Psychometric Society Travel Award , IMPS	<i>2025</i>
Honorable Mention , COMAP Mathematical Contest in Modeling (MCM)	<i>2019</i>
Outstanding professional manager , Biden Consulting Company	<i>2019</i>
Academic Scholarship , Northeastern University	<i>2017–2019</i>
Volunteer Appreciation Award , Children Teaching Program, Paglaum Center	<i>2018</i>
2nd Prize / 3rd Prize , China Undergraduate Mathematical Contest in Modeling	<i>2017–2018</i>
1st Prize / 3rd Prize , NEU Mathematical Modeling Contest for College Students	<i>2017–2018</i>
Outstanding Volunteer , Northeastern University	<i>2017–2018</i>
Outstanding Individual in Social Practice Activities , Northeastern University	<i>2017</i>

Academic Services

Organizing Committee

- 2024 Research Postgraduate Conference (RPC) at HKU

Session Chair

- 2025 National Council on Measurement in Education (NCME) Annual Meeting
- 2024 Research Postgraduate Conference (RPC) at HKU

Reviewer

- 2026 National Council on Measurement in Education (NCME) Annual Meeting
- 2025 National Council on Measurement in Education (NCME) Annual Meeting

Technical Skills

Programming	Python, R, C++, C, MATLAB, SQL
Machine Learning	Scikit-learn, PyTorch, TensorFlow, Keras, OpenCV
Statistical & Research Tools	Mplus, SPSS, LaTeX, RMarkdown, Git, Linux, HTML, CSS, JavaScript

Language Proficiencies

Mandarin	Native
English	Fluent